

DESERTS



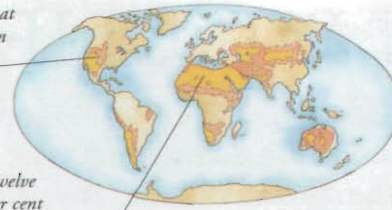
FEW PLACES ON EARTH are as stark and hostile as a desert.

Deserts are vast areas where very little rain falls – typically under

100 mm (3.9 inches) a year. Any rain that does fall quickly evaporates. Few plants can survive, and soil cannot develop in such a dry and barren or arid environment. The landscape is bare sand, gravel, or rock. Clear skies and sparse vegetation leave the ground exposed to extremes of temperature. In the tropics, cloudless skies create hot deserts with daytime temperatures which are often over 50°C (122°F).

Deserts at higher latitudes can be extremely cold.

Areas thought to be at risk of desertification (shown in orange)



Twelve per cent of land is desert.

Desert regions

The world's great deserts lie deep within continents far from the moisture of the oceans. They are also along the Tropics of Cancer and Capricorn, on either side of the Equator, where sinking air creates stable dry weather.

Desert landforms and dunes

Strong winds, sudden flash floods, and exposure to extreme temperatures create distinctive desert features. The wind piles sand up in dunes or sand-blasts rocks. Flash floods carve canyon-like valleys. The desert heat creates corrosive chemicals which sculpt rocks into bizarre shapes.



Oasis

An oasis, such as the Azraq oasis in Jordan (above), is a fertile area within a desert that lies near an underground stream or a spring. Crops such as date palms can grow and desert dwellers can live supported by the land. Artificial oases can be created through irrigation.

Seif dunes form where sand is sparse and wind comes from two directions.

Oasis is a pocket of water.

Butte – eroded mesa

Zeugen rocks are produced by weathering.

Hamada is an area strewn with boulders and stones.

Playa is a dry lake bed of salt.

Eroded arch

Bolson is a drainage basin.

Barchan dune is shaped like a crescent and its tips usually point downwind.

Bajada is a slope of sand deposited by rivers along mountain edges.

Transverse dune – ridge lying across the wind



Mirage

Sometimes the desert heat is so intense that desert travellers believe that they can see water. This is an optical illusion (a trick of the eye), caused by the reflection of a faraway object, which may give the false appearance of a sheet of water.

Types of deserts

Climatic conditions create different types of deserts. In Africa, the Sahara has vast areas of erg (sand seas), hamada (stony plateaus), reg (pebble plains), rocky deserts, canyons, and cliff deserts. In the Antarctic there are ice deserts, while in the deserts of the western USA the heat evaporates rain so quickly that it leaves behind dissolved minerals in a hard, salty crust.



Sandy desert

In flat areas, vast sand seas, or ergs, develop. After a rainfall, water rushes along a wadi, a dry river bed. The sandstone cliffs on either side are gradually worn away by the heat, wind, and rain.



Rocky desert

Many of the world's deserts are strewn with boulders that have been washed there by flash flooding. These rocks are gradually broken down by the action of wind and weather.

Rocks brought to desert by flood water.

The shifting desert

As climatic conditions change, deserts shrink and expand. In the past, the Sahel, the southern margin of the Sahara, was watered by summer rains moving up from the south. In recent years, the lack of rain in the Sahel has caused drought and famine in places such as Sudan and Ethiopia.

Desertification

The effect of drought and heavy grazing by cattle, sheep, and goats destroys vegetation cover, turning the area permanently to desert. This process is known as desertification.



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